

Aircraft Fuzzy Control System

Abstract

This report presents the design and implementation of a fuzzy logic-based control system for aircraft behavior across different flight phases. The system uses linguistic rules and membership functions to control outputs like thrust and elevator deflection. It features an explanation engine, emergency fail-safe logic, and logs every simulation to a file.

1. Introduction

Aircraft control systems often encounter uncertainty due to dynamic and nonlinear environments. Classical control methods may fail in such situations. Fuzzy logic offers a robust solution by allowing rules based on expert knowledge and linguistic variables. This project proposes a modular fuzzy control system tailored to each flight phase.

2. System Overview

The fuzzy control system is divided into five flight phases:

- Takeoff
- Climb
- Cruise
- Approach
- Landing

Each phase has its own dedicated `.fis` file. A fuzzy inference engine evaluates input conditions and determines the appropriate control outputs. An explanation module provides interpretability, while a fail-safe mechanism handles emergencies.

3. Fuzzy Logic Design

To ensure high performance and precision during every stage of flight, the system employs five dedicated Fuzzy Inference Systems (FIS), each tailored to a specific flight phase:

- `plane_takeoff.fis`
- `plane_climb.fis`
- `plane_cruise.fis`
- `plane_approach.fis`
- `plane_landing.fis`

Each FIS operates using Mamdani-type fuzzy logic, which is well-suited for control systems requiring human-like reasoning, interpretable rule sets, and smooth transitions between actions. This design enables the controller to mimic real pilot decision-making processes in a structured, explainable way.

Each phase of flight presents distinct control challenges. Rather than relying on a single, overly complex controller to handle all scenarios, the system breaks the task into five specialized FIS controllers, each focused on the operational context of one phase:

1. **Takeoff (`plane_takeoff.fis`)**
 - Prioritizes safe lift-off, stable acceleration, and thrust balance.
 - Handles low-speed dynamics, engine performance, and initial pitch control.
2. **Climb (`plane_climb.fis`)**
 - Focuses on altitude gain, engine stress management, and climb rate control.
 - Balances fuel efficiency with climb aggressiveness.
3. **Cruise (`plane_cruise.fis`)**
 - Maintains altitude, optimizes fuel usage, and monitors engine health over long distances.
 - Handles turbulence response and passenger comfort stabilization.
4. **Approach (`plane_approach.fis`)**
 - Prepares aircraft for descent, ensuring speed reduction, altitude precision, and conflict resolution.
 - Engages stall detection, obstacle avoidance, and stress reduction features.
5. **Landing (`plane_landing.fis`)**
 - Manages critical final moments of flight: gear deployment, safe descent, and touchdown control.
 - Uses fine-tuned control of elevator and thrust to ensure a safe and smooth landing.

The heart of each FIS is a rule base composed of over 2,000 handcrafted fuzzy rules, developed from aviation domain expertise, safety procedures, and pilot behavior models. These rules take combinations of input variables such as airspeed, altitude, turbulence, fuel status and map them to control decisions like:

- “If **airspeed is low** and **pitch angle is nose up**, then **stall warning is high**”
- “If **engine health is degraded** and **fuel is low**, then **suggest emergency diversion**”

Each rule is linguistically interpretable and can be traced to real-world pilot responses under various conditions. The system is also designed to allow adaptive switching between these FIS blocks based on current flight parameters, enabling continuous, context-sensitive control.

4. Input and Output Variables

The fuzzy control system is designed to evaluate and respond to a variety of critical parameters during different phases of flight. It utilizes 9 input variables and generates 6 output commands, allowing it to handle complex and dynamic flight situations with high adaptability.

Input Variables:

- **Turbulence:** Indicates the severity of atmospheric disturbances affecting the aircraft. This input helps the system adjust control surfaces and autopilot logic to maintain stability and reduce stress during turbulence.
- **Air speed:** Measures the speed of the aircraft relative to the air. It is a fundamental parameter for detecting stall risks, adjusting thrust, and ensuring safe flight envelopes are maintained.
- **Altitude:** Represents the aircraft’s vertical position above sea level. It influences engine behavior, autopilot decisions, and pressure-related performance adjustments.
- **Pitch angle:** The angular orientation of the aircraft’s nose relative to the horizon. This is critical for understanding climb, descent, or level flight and detecting abnormal attitudes.
- **Vertical speed:** The rate at which the aircraft is ascending or descending. It helps the system react to rapid altitude changes and smooth transitions between flight phases.
- **Engine health:** Reflects the current condition of the engines (optimal, degraded, or critical). This input is essential for fail-safe decisions, fuel management, and emergency thrust control.

- **Fuel status:** Indicates remaining fuel percentage. It is vital for determining energy management strategies, economic cruising, or activating emergency diversion logic.
- **Proximity to obstacles:** Measures how close the aircraft is to terrain or external hazards. This enhances safety by supporting collision avoidance and resolution advisory maneuvers.
- **TCAS conflict level:** Derived from the Traffic Collision Avoidance System. It indicates potential conflict with other aircraft and directly influences autopilot override and evasive action logic.

Output Variables:

- **Thrust command:** Sets the desired engine thrust level. It is affected by speed, altitude, and emergency needs to maintain proper energy states or respond to failures.
- **Elevator deflection:** Determines how much the elevator should move to control pitch. This directly affects climb, descent, and stabilization under turbulence or stall.
- **Autopilot mode suggestion:** Advises on the optimal autopilot mode (e.g., manual, altitude hold, approach). The system can override modes for safety or return to automation after correction.
- **Stall warning:** Activates if stall conditions are predicted. It serves as a proactive safety signal to inform the pilot or automated system of imminent aerodynamic risk.
- **Fuel reserve command:** Suggests fuel usage strategy (normal, economy, or emergency). This output helps conserve fuel or initiate diversion if reserves are critically low.
- **Stress reduction aid:** Proposes automatic adjustments (like increasing autopilot assistance or reducing maneuvers) to lower pilot workload and structural stress during challenging conditions.

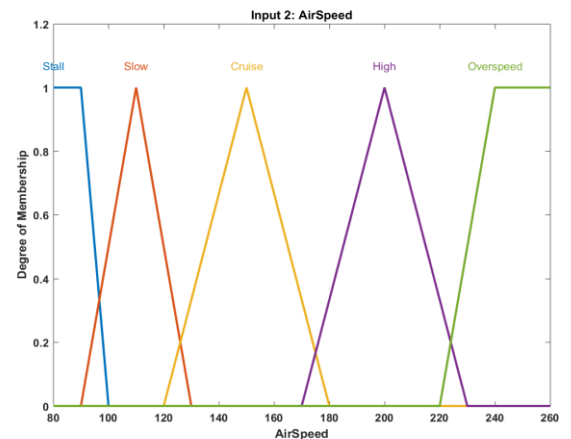
Each input and output variable are chosen with direct relevance to flight safety, efficiency, and adaptability. The fuzzy logic system uses these parameters to model expert pilot behavior and ensure resilient aircraft control across all flight phases.

5. Membership Functions

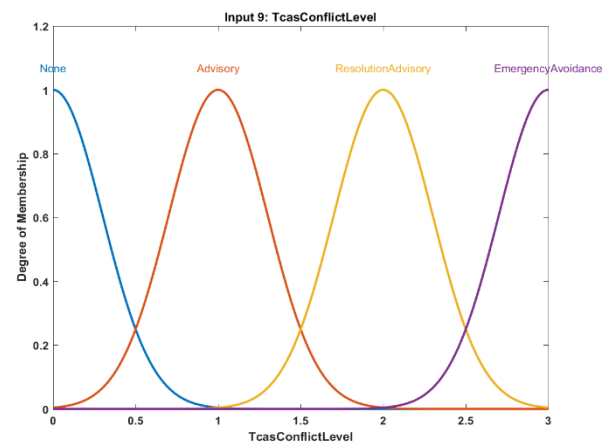
Each variable is described using fuzzy sets (linguistic terms) with triangular, trapezoidal, or Gaussian membership functions.

Example:

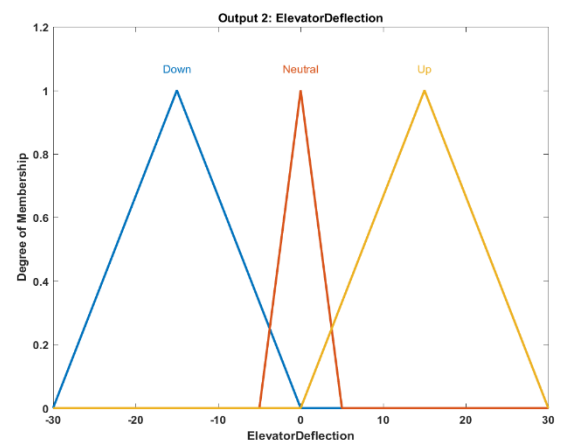
- **Air Speed** includes: *stall*, *slow*, *cruise*, *high*, *overspeed*



- **TCAs Conflict Level** includes: *none*, *advisory*, *resolution advisory*, *emergency avoidance*



- **Elevator Deflection** includes: *down*, *neutral*, *up*



Membership functions were chosen to reflect realistic boundaries and overlapping conditions in flight behavior.

6. Emergency and Fail-Safe Logic

In critical or hazardous flight conditions, the system overrides standard fuzzy inference and immediately activates hard-coded emergency responses. This fail-safe mechanism ensures rapid and deterministic action when safety thresholds are breached.

Emergency control is triggered when any of the following conditions are met:

- **Stall risk:** Airspeed < 100 km/h and Pitch angle $> 10^\circ$
- **Combined Fuel and Engine Degradation:** Fuel status $< 10\%$ and Engine health < 1.0
- **Collision Threat:** Proximity to obstacles < 150 m and TCAS conflict level ≥ 2
- **Severe Turbulence:** Turbulence > 2.5

When a fail-safe condition is detected, the system enforces the following safety actions:

- **Thrust command** is set to maximum
- **Elevator deflection** is forced downward
- **Autopilot mode** switches to manual
- A clearly formatted **fail-safe warning box** is printed to alert the pilot or system operator

This mechanism guarantees immediate corrective action during emergency conditions, bypassing fuzzy logic uncertainty to protect aircraft stability, mission integrity, and passenger safety.

7. MATLAB Implementation

The system is built in **MATLAB R2015** with Fuzzy Logic Toolbox.

Key Files:

- `interactive_fuzzy_explainer.m`: UI, evaluation, explanation, and logging
- `explain_from_cleaned_file.m`: rule matching and retrieval

- `plot_all_mfs.m`: plots all membership functions
- `expln_rules_cleaned_final.txt`: cleaned expert rules with explanation

8. Sample Scenario and Output

Flight Phase: **Approach**

Input name	Value	Label	Degree
Turbulence	2.0	Severe	0.61
Air speed	90.00	Stall	1.00
Altitude	2000.00	Low	1.00
Pitch angle	12.00	Nose up	0.70
Vertical speed	-500.00	Level	0.50
Engine health	0.50	Optimal	0.25
Fuel status	5.00	Empty	1.00
Proximity to obstacles	80.00	Immediate danger	0.40
TCA's conflict level	3.00	Emergency avoidance	1.00

Output name	Value	Label	Degree
Thrust command	100.00	Max	1.00
Elevator deflection	-20.00	Down	1.00
Autopilot mode suggestion	0.00	Manual	1.00
Stall warning	1.00	Warning	1.00
Fuel reserve command	0.00	Emergency diversion	1.00
Stress reduction command	1.00	Override manual	1.00

Explanation
- Low airspeed and pitch attitude indicate high stall risk; issuing stall advisory is critical. - Vertical dynamics suggest aggressive altitude change, requiring corresponding thrust and control. - Engine health degraded; thrust adjusted to ensure performance under failure conditions. - High TCAS conflict level; autopilot mode and manual override may be prioritized for safety.

Main Contributing Factors		
Air speed	90.00	Stall
Altitude	2000.00	Low
Fuel status	5.00	Empty
TCAs conflict level	3.00	Emergency avoidance

FAIL SAFE MODE ACTIVATED

A critical condition was detected
Emergency control outputs were issued.
Fuzzy logic system was overridden for safety response.

9. Output Logging System

Each run:

- Generates .txt log file
- Logs inputs, outputs, degrees, labels
- Captures explanation and emergency flags
- Helps in post-analysis and auditing

Example log file: [log_2025_06_08_14_22_58.txt](#)

13. Conclusion

The aircraft fuzzy control system effectively models expert decision-making across all flight phases. Its structured design, explanation engine, emergency logic, and clear input/output representation make it a strong base for real-world embedded control applications.